

Math 372: Homework 08

Due Friday, April 4

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1 (Central Limit Theorem). One hundred 6-sided dice are rolled, and their numbers are added up. Find the probability that the sum of the numbers on the dice is between 330 and 380.

Problem 2. The new Pokemon TCG expansion set *Journey Together* just came out on Friday, March 28. The *Journey Together* expansion set consists of 190 distinct cards. Cards can be purchased in packs consisting of 10 random cards. Assume that all cards are equally likely to appear in a pack. How many packs do you need to buy to ensure that the probability that you have purchased all 190 cards is at least 95%? Of course, the packs are \$6 each, and we want to spend as little money as possible, so find the *smallest* number of packs that you need to buy.

Hint: Adapt the argument presented for the “coupon collector” problem from lecture 24.

Extra credit: write a computer simulation to answer this problem.

Problem 3. Annie and Alvie have agreed to meet for lunch between noon (0:00 pm) and 1:00pm. Denote Annie's arrival time by X and Alvie's by Y . Suppose that X and Y are independent with pdfs

$$f_X(x) = \begin{cases} 3x^2 & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise} \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} 2y & : 0 \leq y \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

What is the expected amount of time that the one who arrives first must wait for the other person. Hint: $h(X, Y) = |X - Y|$

Problem 4. The joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} \frac{27y^2 e^{-3x}}{x^2} & : 0 < y < x < \infty \\ 0 & : \text{otherwise} \end{cases}$$

(a) Find the marginal density for X .

(b) Are X and Y independent? Why or why not?

(c) Find $\mathbb{P}[X > 2Y]$

Problem 5. The density function for the standard normal distribution is the function

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

Show that ϕ is concave down for $|z| < 1$, concave up for $|z| > 1$, with inflection points at $z = \pm 1$.

Problem 6. Let (X, Y) be uniformly distributed on the unit square $0 < x < 1$, $0 < y < 1$. Find the following probabilities:

(a) $\mathbb{P}[X < Y]$

(b) $\mathbb{P}[X + Y < 1]$

(c) $\mathbb{P}[\min(X, Y) > \frac{3}{4}]$

(d) $\mathbb{P}[XY \leq \frac{1}{2}]$

Problem 7 (Extra credit). An urn contains one red ball and one blue ball. A ball is drawn at random. If it is red, the game is over. If it is blue, it is replaced in the urn along with an extra blue ball. This process continues until the red ball is drawn, however long it takes. Let Y be the number of draws required.

(a) Argue that $\mathbb{P}[Y = k] = \frac{1}{k(k+1)}$

(b) Show that this defines a valid pmf. (Hint: first show that $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, and then consider $\sum_{k=1}^n \frac{1}{k(k+1)}$).

(c) Show that $\mathbb{E}[Y]$ does not exist.

HW 8 Solutions

1. Let X_i = roll of i^{th} dice.

We want $P[330 \leq X_1 + \dots + X_{100} \leq 380]$.

or equivalently, $P[3.3 \leq \frac{X_1 + \dots + X_{100}}{100} \leq 3.8]$.

By Central Limit Theorem,

$$\frac{X_1 + \dots + X_{100}}{100} \approx \mu + \frac{\sigma}{\sqrt{100}} Z$$

where $Z \sim N(0, 1)$, $\mu = E[X_i] = 3.5$,

and $\sigma^2 = \text{Var}(X_i) = E[X_i^2] - (E[X_i])^2$

$$= \frac{1+4+9+16+25+36}{6} - (3.5)^2 = \frac{35}{12}$$

So $\sigma = \sqrt{\frac{35}{12}} \approx 1.7$

$$= \frac{1+4+9+16+25+36}{6} - (3.5)^2 = \frac{35}{12}$$

This should be $\frac{1}{10} \sqrt{\frac{35}{12}} Z$

$$\frac{X_1 + \dots + X_{100}}{100} \approx 3.5 + \frac{35}{120} Z$$

$$\text{So } P[3.3 \leq \frac{X_1 + \dots + X_{100}}{100} \leq 3.8] \approx P[3.3 \leq 3.5 + \frac{35}{120} Z \leq 3.8]$$

$$= P[-0.2 \leq \frac{35}{120} Z \leq 0.3] = P[-0.69 \leq Z \leq 1.029]$$

$$\approx 0.6$$

So there is about 60% chance of the sum of the dice being between 330 and 380.

$$P\left[3.3 \leq \frac{X_1 + \dots + X_{100}}{100} \leq 3.8\right]$$

$$\approx P\left[3.3 \leq 3.5 + \frac{1}{10} \sqrt{\frac{35}{12}} Z \leq 3.8\right]$$

$$= P\left[-0.2 \leq \frac{1}{10} \sqrt{\frac{35}{12}} Z \leq 0.3\right]$$

$$= P\left[-2 \leq \sqrt{\frac{35}{12}} Z \leq 3\right]$$

$$= P\left[\frac{-2\sqrt{12}}{\sqrt{35}} \leq Z \leq \frac{3\sqrt{12}}{\sqrt{35}}\right]$$

$$\approx P[1.17 \leq Z \leq 1.76]$$

$$\approx 0.84$$

#2 Let $m \geq 1$. Let's denote the cards by $1, 2, \dots, 190$.

Let $A_k = [\text{card } k \text{ isn't in the first } m \text{ packs}]$.

Let $V =$ the number of packs purchased until every card is purchased at least once.

Now, $P[A_k] = \left(1 - \frac{1}{190}\right)^{10m}$ for every $k \in \{1, \dots, 190\}$.

Hence, $P[V > m] = P[\text{At least one card is missing from first } m \text{ packs}]$

$$= P[A_1 \cup \dots \cup A_{190}]$$

$$\leq P[A_1] + \dots + P[A_{190}]$$

$$\leq 190 \left(1 - \frac{1}{190}\right)^{10m}$$

$$= 190 \left[\left(1 - \frac{1}{190}\right)^{190}\right]^{\frac{10m}{190}}$$

$$\approx 190 [e^{-1}]^{\frac{m}{19}}$$

$$= 190 e^{-m/19}$$

(Since $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}$)

We need m to be large enough that $P[V \leq m] \approx 0.95$, or equivalently that $P[V > m] \approx 0.05$. By the above calculation, it suffices for $190 e^{-m/19} \approx 0.05$. Solving for m gives

$m = 156.6$. So we need to buy at least 157 packs to have probability $\geq 95\%$ of catching all the pokemon. (That's \$942 + tax!)

$$(3) E[|X-Y|] = \int_0^1 \int_0^1 |x-y| f_X(x) f_Y(y) dx dy$$

$$= \int_0^1 \int_0^1 |x-y| (3x^2)(2y) dx dy$$

$$= 6 \int_0^1 y \left[\int_0^1 |x-y| x^2 dx \right] dy \quad (*)$$

Call it I

$$I = \int_0^y (y-x) x^2 dx + \int_y^1 (x-y) x^2 dx$$

$$= \int_0^y (yx^2 - x^3) dx + \int_y^1 (x^3 - yx^2) dx$$

$$= \left[\frac{x^3 y}{3} - \frac{x^4}{4} \right]_{x=0}^{x=y} + \left[\frac{x^4}{4} - \frac{x^3 y}{3} \right]_{x=y}^{x=1}$$

$$= \left(\frac{y^4}{3} - \frac{y^4}{4} \right) - (0 - 0) \\ + \left(\frac{1}{4} - \frac{y}{3} \right) - \left(\frac{y^4}{4} - \frac{y^4}{3} \right)$$

$$= \frac{2}{3}y^4 - \frac{y^4}{2} + \frac{1}{4} - \frac{y}{3}$$

$$= \frac{y^4}{6} - \frac{y}{3} + \frac{1}{4}$$

Plug back into (*):

$$E[|X-Y|] = 6 \int_0^1 y \left[\frac{y^4}{6} - \frac{y}{3} + \frac{1}{4} \right] dy$$

$$= 6 \int_0^1 \left(\frac{y^5}{6} - \frac{y^2}{3} + \frac{y}{4} \right) dy$$

$$= 6 \left[\frac{y^6}{36} - \frac{y^3}{9} + \frac{y^2}{8} \right]_0^1 = 6 \left(\frac{1}{36} - \frac{1}{9} + \frac{1}{8} \right)$$

$$= 6 \left(\frac{1}{24} \right) = \frac{1}{4} = \boxed{.25}$$

Expected wait is 15 min

$$\textcircled{4} \textcircled{a} f_x(x) = \int_{\text{all } y} f(x, y) dy$$

$$= \int_0^x f(x, y) dy \quad (\text{since } 0 < y < x)$$

$$= \int_0^x \frac{27y^2 e^{-3x}}{x^2} dy$$

$$= \frac{e^{-3x}}{x^2} \int_0^x 27y^2 dy$$

$$= \frac{e^{-3x}}{x^2} \left[9y^3 \right]_0^x$$

$$= \frac{e^{-3x}}{x^2} [9x^3 - 0]$$

$$= \boxed{9e^{-3x} x}$$

for $x > 0$

④ (b) No. If they were indep, we could write

$$f(x, y) = f_X(x) f_Y(y), \text{ where } f_X \text{ depends only on } x \text{ and } f_Y \text{ depends only on } y.$$

But, Using $f(x, y) = \frac{27y^2 e^{-3x}}{x^2} \mathbb{1}_{[0 < y < x]}$

$$\text{and } f_X(x) = 9x e^{-3x} \mathbb{1}_{[x > 0]},$$

We have

$$f_Y(y) = \frac{f(x, y)}{f_X(x)} = \frac{3y^2}{x^3} \mathbb{1}_{[0 < y < x]},$$

a contradiction since the Left hand side ~~is~~ depends on x but the right hand side must not.

(^{other} Heuristic arguments are fine here).

$$(c) \quad P[X > 2Y] = P[Y < \frac{X}{2}]$$

$$= \int_0^{\infty} \int_0^{x/2} f(x, y) dy dx$$

$$= \int_0^{\infty} \int_0^{x/2} \frac{27y^2 e^{-3x}}{x^2} dy dx$$

$$= \int_0^{\infty} \frac{e^{-3x}}{x^2} \left[\int_0^{x/2} 27y^2 dy \right] dx$$

$$= \int_0^{\infty} \frac{e^{-3x}}{x^2} [9y^3]_0^{x/2} dx$$

$$= \int_0^{\infty} \frac{e^{-3x}}{x^2} 9 \left(\frac{x}{2} \right)^3 dx = \frac{9}{8} \int_0^{\infty} x e^{-3x} dx$$

$$= \frac{9}{8} \int_0^{\infty} x \frac{d}{dx} \left[\frac{e^{-3x}}{-3} \right] dx = \frac{9}{8} \left\{ \frac{-x e^{-3x}}{3} \Big|_0^{\infty} + \int_0^{\infty} \frac{e^{-3x}}{3} dx \right\}$$

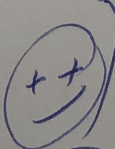
$$= \frac{3}{8} \int_0^{\infty} e^{-3x} dx = \frac{3}{8} \left[\frac{e^{-3x}}{-3} \right]_0^{\infty} = \boxed{\frac{1}{8}}$$

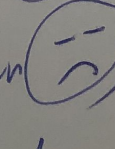
$$(5) \quad \phi'(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \underbrace{\frac{d}{dz} \left[-\frac{z^2}{2} \right]}_{-\frac{2z}{2} = -z} \quad (\text{Chain Rule})$$

$$= -\frac{z}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\begin{aligned} \phi''(z) &= -\frac{1}{\sqrt{2\pi}} e^{-z^2/2} + \left(-\frac{z}{\sqrt{2\pi}} \right) e^{-z^2/2} (-z) \quad (\text{Product Rule, Chain Rule}) \\ &= (z^2 - 1) \frac{e^{-z^2/2}}{\sqrt{2\pi}} \end{aligned}$$

Thus, $\phi''(z) = 0$ iff $z = \pm 1$, ~~these~~

~~and~~ $\phi''(z) > 0$ iff $|z| > 1$ (concave up )

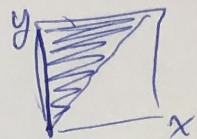
$\phi''(z) < 0$ iff $|z| < 1$ (concave down )

Concavity of the function is determined by 2nd derivative.

Inflection points are where 2nd deriv switches from $\oplus$ to $\ominus$ or from $\ominus$ to $\oplus$

⑥ Observe $f(x,y) = 1$ for all $x,y \in (0,1)$.

① $P[X < Y] = \int_0^1 \int_0^1 1_{[x < y]} f(x,y) dx dy$



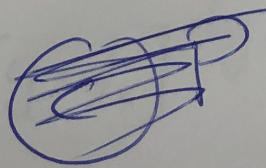
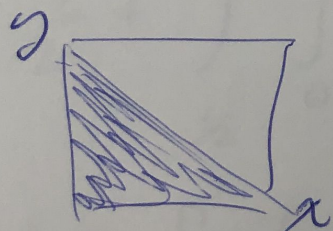
$$= \int_0^1 \int_0^y dx dy = \int_0^1 y dy = \left[\frac{y^2}{2} \right]_0^1 = \boxed{\frac{1}{2}}$$

Also, $\frac{1}{2}$ makes sense by symmetry.

② $P[X+Y < 1] = \int_0^1 \int_0^1 1_{[x+y < 1]} dx dy$

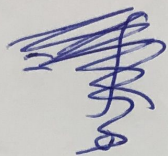
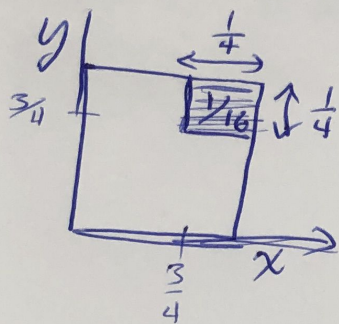
$$= \int_0^1 \int_0^{1-y} dx dy = \int_0^1 (1-y) dy$$

$$= \left[y - \frac{y^2}{2} \right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$$

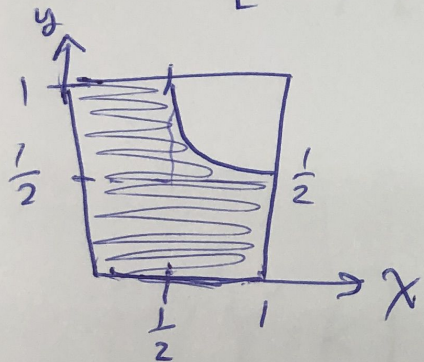


$$(c) P[\min(X, Y) > \frac{3}{4}] = P[X > \frac{3}{4} \text{ and } Y > \frac{3}{4}]$$

$$= P[X > \frac{3}{4}] P[Y > \frac{3}{4}] = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$



$$(d) P[XY < \frac{1}{2}] = \int_0^1 \int_0^1 \mathbb{1}_{[xy < \frac{1}{2}]} dy dx$$



$$= \underbrace{\int_0^{1/2} \int_0^1 \mathbb{1}_{[xy < \frac{1}{2}]} dy dx}_{= \frac{1}{2}} + \int_{1/2}^1 \int_0^{1/(2x)} \mathbb{1}_{[xy < \frac{1}{2}]} dy dx$$

$$= \frac{1}{2} + \int_{1/2}^1 \int_0^{1/(2x)} \mathbb{1}_{[y < \frac{1}{2x}]} dy dx = \frac{1}{2} + \int_{1/2}^1 \int_0^{1/(2x)} dy dx$$

$$= \frac{1}{2} + \int_{1/2}^1 \frac{1}{2x} dx = \frac{1}{2} + \frac{1}{2} [\log(x)]_{1/2}^1$$

$$= \frac{1}{2} + \frac{1}{2} (-\log(\frac{1}{2})) = \frac{1}{2} + \frac{1}{2} \log(2) \approx \boxed{0.85}$$

~~Let $X = 1, 2, \dots$~~
~~if the ball is BLUE~~
~~if the ball is RED~~

~~For each $k = 1, 2, \dots$~~

~~$P[Y=1]$~~ $P[Y=1] = \frac{1}{2}$

$$P[Y=2] = \frac{1}{2} \cdot \frac{1}{3}$$

$$P[Y=3] = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{3} \cdot \frac{1}{4}$$

$$P[Y=4] = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5} = \frac{1}{4} \cdot \frac{1}{5}$$

first 3
balls are
blue

last ball
is red, given first 3
are blue

$$P[Y=k] = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \left(\frac{k-1}{k}\right) \cdot \frac{1}{k+1} = \frac{1}{k(k+1)}$$

first $k-1$ balls are
blue

k^{th} ball
is red, given
prior balls blue

(b) $\sum_{k=1}^{\infty} P[Y=k] = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \left[\frac{1}{k} - \frac{1}{k+1} \right]$

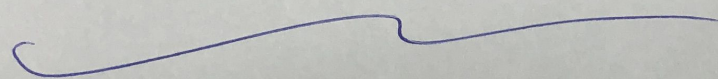
$$= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} + \dots$$

$$= 1$$

$$(c) \quad E[Y] = \sum_{k=1}^{\infty} k P[Y=k]$$

$$= \sum_{k=1}^{\infty} k \cdot \frac{1}{k(k+1)} = \sum_{k=1}^{\infty} \frac{1}{k+1}$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = +\infty.$$



Harmonic Series

So $E[Y]$ is NOT FINITE.